

Essay practice — draw and/or explain

7 essays × 5 pts · answer by drawing, explaining or both · rubrics and model answers start on a new page

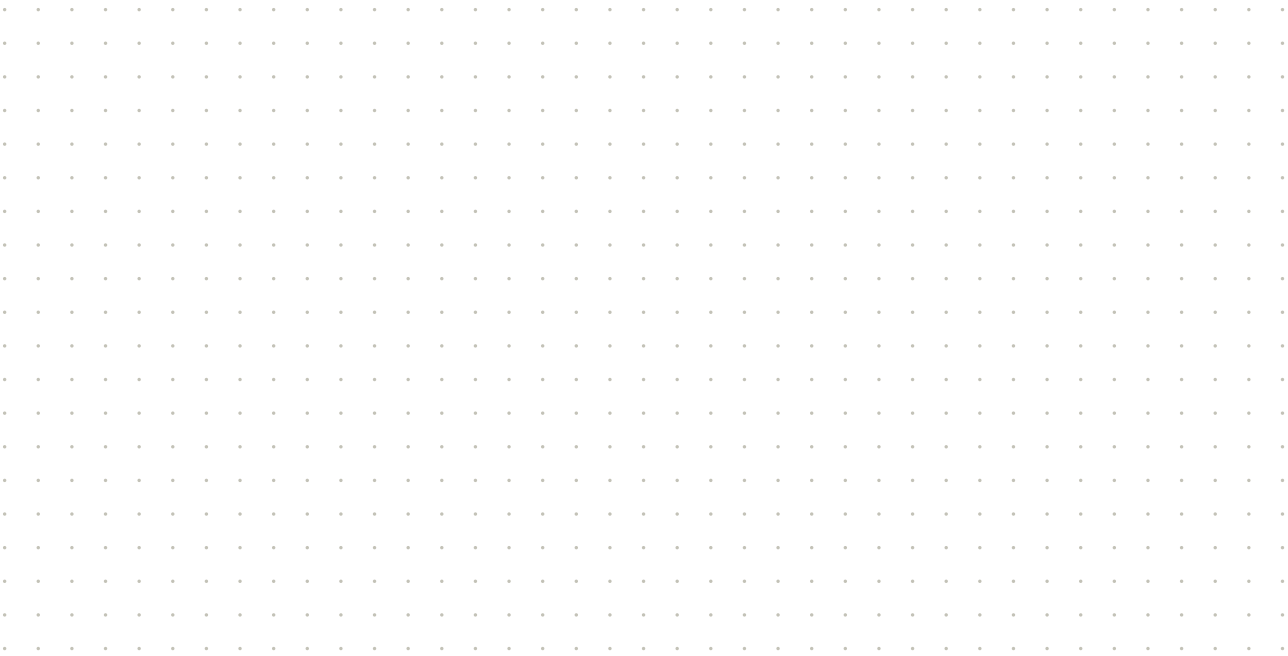
Answer in the space provided — draw, explain, or both.

- 1. Compare **LAN**, **WAN** and **wireless** networks: what each one covers, the media it typically uses, and examples [5 pts]
of each. Sketch the three side by side if it helps. You may draw, explain, or both.

DRAW AND/OR EXPLAIN

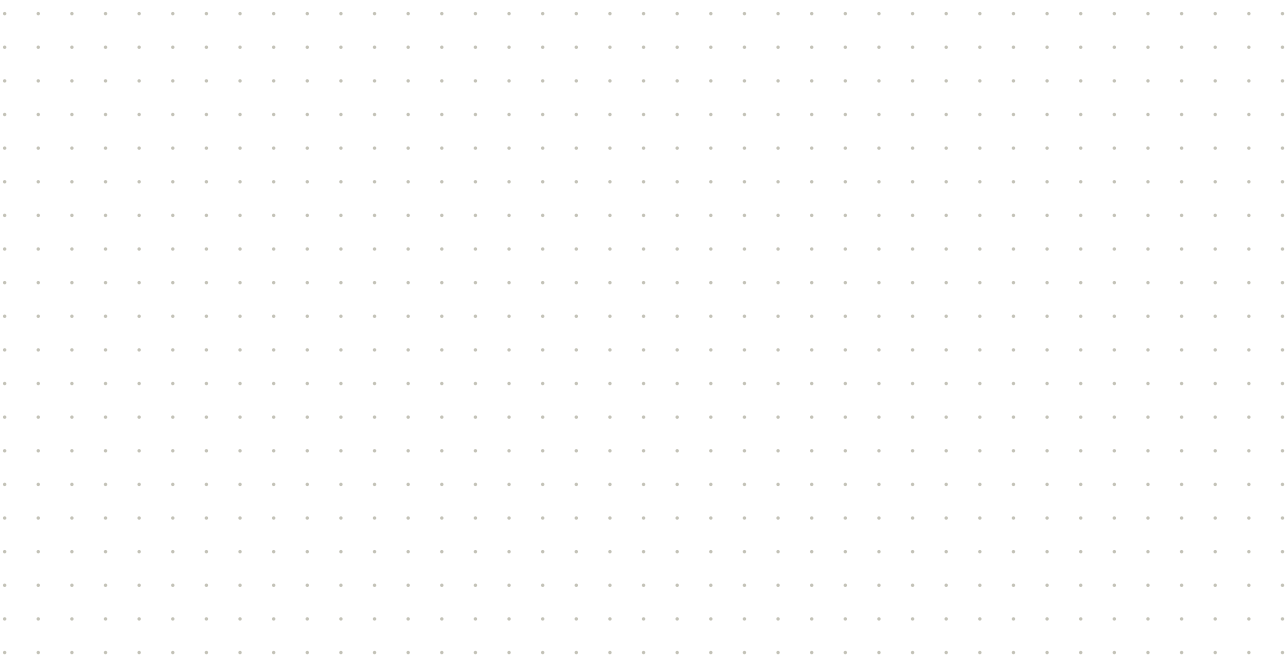
2. Draw a **composite signal** made of two sine waves, of frequency f and $3f$, in the **time domain**, then draw its **frequency-domain** spectrum. Define **bandwidth** and work it out for the slide example whose components run from 1000 Hz to 5000 Hz. You may draw, explain, or both. [5 pts]

DRAW AND/OR EXPLAIN



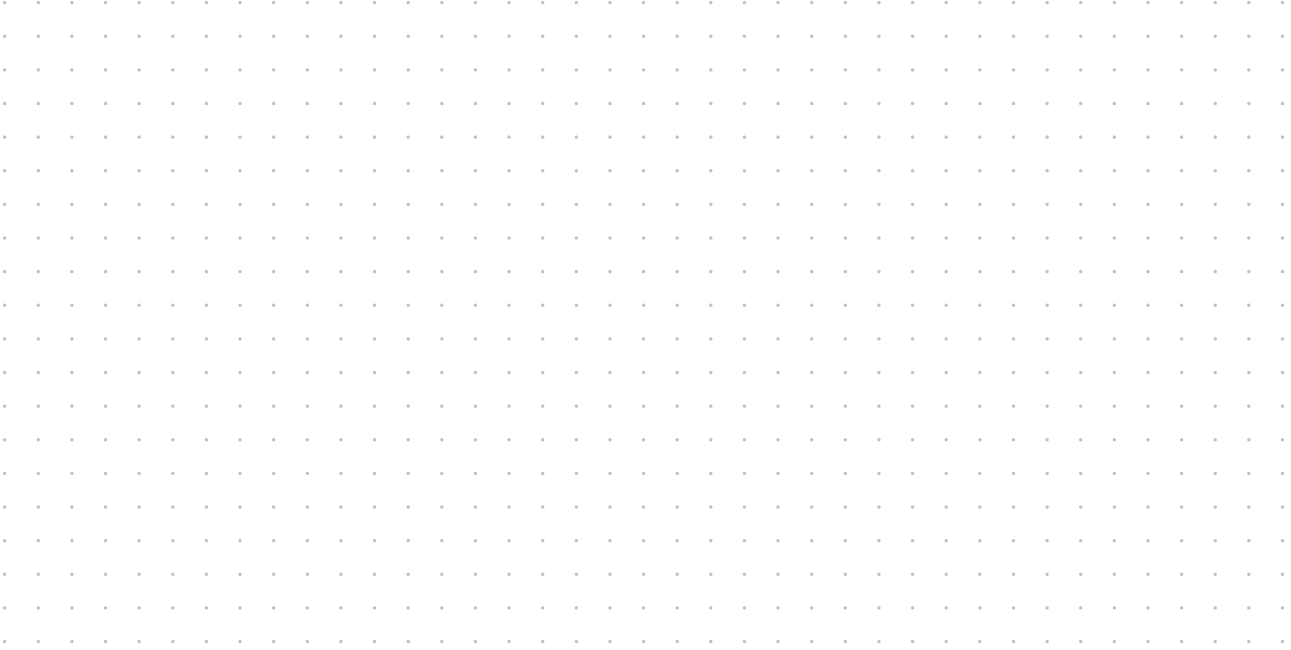
3. Compare **baseband** and **broadband** transmission. Draw a **low-pass** channel and a **bandpass** channel on a frequency axis, then explain when each kind of transmission is used. You may draw, explain, or both. [5 pts]

DRAW AND/OR EXPLAIN



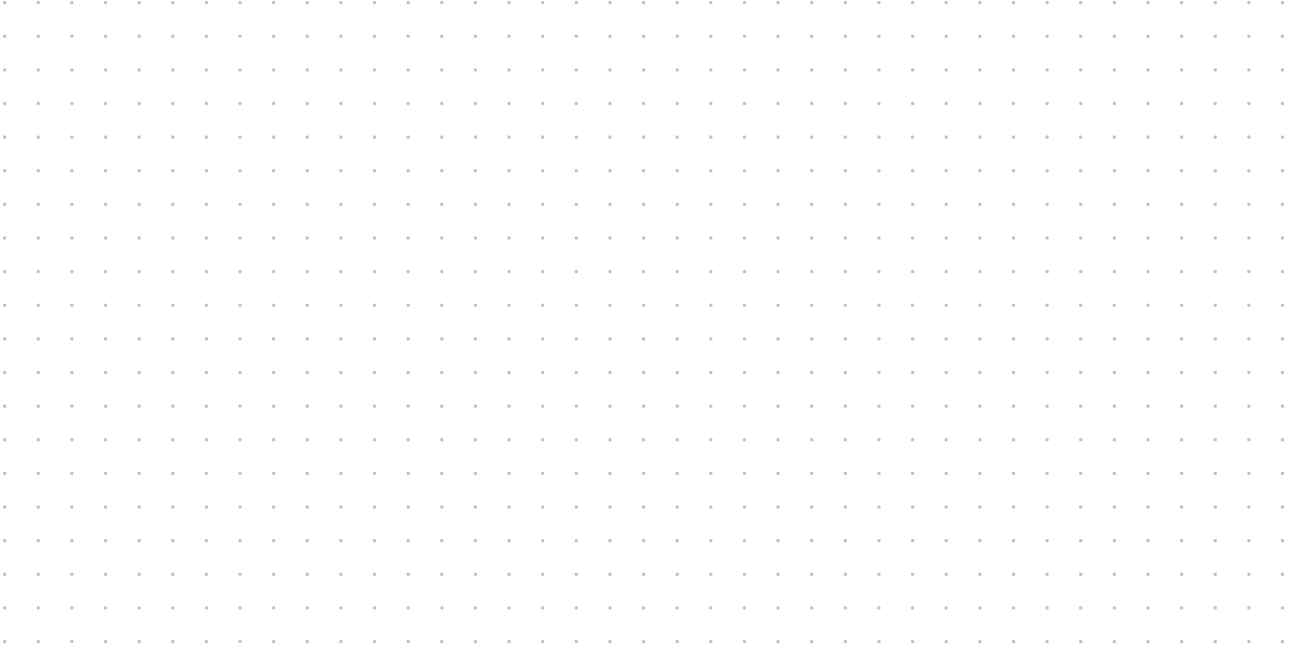
4. Draw the **block coding** process for **4B/5B** and explain its three steps. Say why 4B/5B is paired with **NRZ-I**, what it does to the bit rate, and how block coding differs from **scrambling**. You may draw, explain, or both. [5 pts]

DRAW AND/OR EXPLAIN



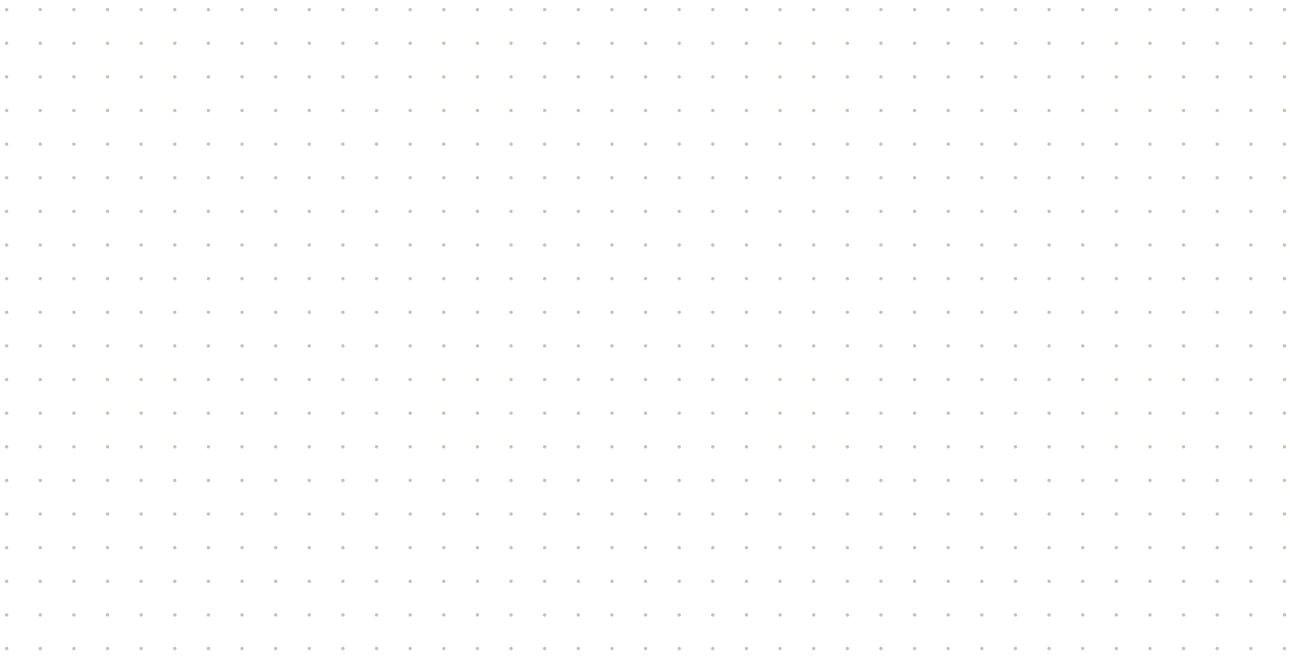
5. Compare **delta modulation** (DM) with **PCM**. Draw a short DM staircase together with the bits it sends, say what DM sends for each sample, and explain the trade-off. You may draw, explain, or both. [5 pts]

DRAW AND/OR EXPLAIN



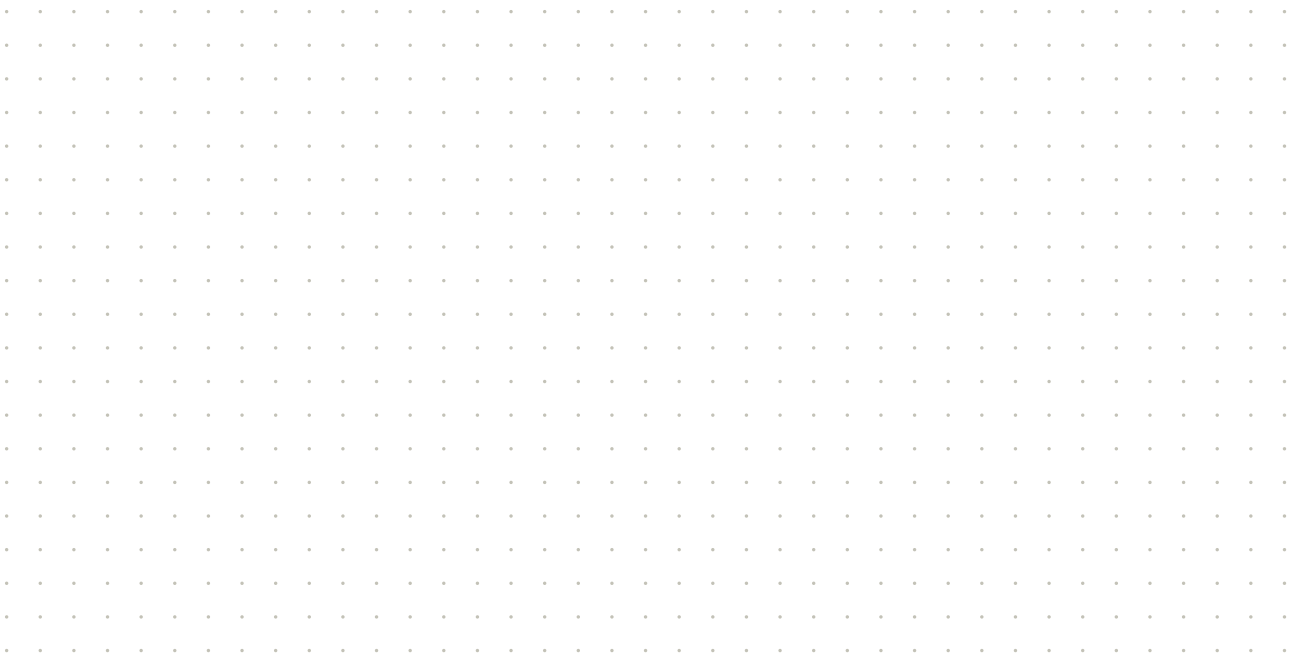
6. Compare **parallel** and **serial** transmission, and the three kinds of serial transmission: **asynchronous**, **synchronous** and **isochronous**. Draw an asynchronous frame, labelling the start and stop bits. You may draw, explain, or both. [5 pts]

DRAW AND/OR EXPLAIN



7. Draw the **QPSK** constellation with the slides' bit labels and explain what a constellation diagram shows. Then explain how **16-QAM** carries more bits per signal element. [5 pts]

DRAW AND/OR EXPLAIN



Rubrics and model answers

1. ESSAY

Compare **LAN**, **WAN** and **wireless** networks: what each one covers, the media it typically uses, and examples of each. Sketch the three side by side if it helps. You may draw, explain, or both.

RUBRIC

1 pt LAN coverage: a connection of computers typically within 5 km, with the computers normally adjacent to each other.

1 pt LAN media and standards: the computers are interconnected using copper or fiber media, and a LAN typically uses the IEEE 802 series (e.g. 802.3 Ethernet); the IEEE took over LAN standards for the physical and data link layers.

1 pt WAN: geographically long-distance connections — a long cable, normally fiber optic (dark fiber = unused fiber; examples: submarine cable, fiber laid beside roads or railways), or microwave through satellites (e.g. Starlink) and terrestrial microwave links.

1 pt Wireless: unguided media using radio signals (also microwave) over a short or long distance; it can be a personal, local or wide area network and uses a “cell” to interconnect mobile devices (examples: Bluetooth, the IEEE 802.11 series, the LTE mobile phone network).

1 pt A clear labelled sketch or comparison table that puts the three types side by side (coverage, medium, example) and shows the wireless network without a cable.

MODEL ANSWER

The slides sort the networks of the data link layer into three types: LAN, WAN and wireless. L01 p6

Type	What it covers	Typical media	Examples
LAN	Computers typically within 5 km , normally adjacent to each other L01 p7	Copper or fiber ; typically the IEEE 802 series L01 p7	Ethernet (IEEE 802.3, CSMA/CD) L01 p9
WAN	Geographically long-distance connections L01 p6	A long cable, normally fiber optic (dark fiber = unused fiber); microwave through satellites and terrestrial links L01 p10	Submarine cable; fiber laid beside roads or railways; satellites such as SpaceX Starlink (550 km above the earth); terrestrial microwave link L01 p10 L01 p11
Wireless	Short or long distance; can be a personal, local or wide area network L01 p6 L01 p12	Unguided media: radio signals, also microwave; a “cell” interconnects mobile devices L01 p12	Bluetooth; IEEE 802.11 series; LTE mobile phone network L01 p12

If drawn: a LAN as a few computers joined by copper or fiber; a WAN as distant sites joined by a long fiber, a submarine cable and a satellite with its uplink and downlink; a wireless network as mobile devices linked through a cell, with no cable between them. L01 p10 L01 p12

The IEEE took over LAN standards — primarily for the physical and data link layers, using the OSI model as a framework — after an early period of proprietary, vendor-locked networks. L01 p8 Of the 802 numbers, 802.3 is CSMA/CD Ethernet and 802.11 is wireless. L01 p9

The types overlap: “wireless” describes the medium (unguided), and a wireless network can itself be a personal, local or wide area network. L01 p12 If the terrestrial microwave link is drawn, show it as a line-of-sight path; the slide’s picture of ground and sky waves is wrong. L01 p10

2. ESSAY

Draw a **composite signal** made of two sine waves, of frequency f and $3f$, in the **time domain**, then draw its **frequency-domain** spectrum. Define **bandwidth** and work it out for the slide example whose components run from 1000 Hz to 5000 Hz. You may draw, explain, or both.

RUBRIC

1 pt Composite signal: a combination of simple sine waves with different frequencies, amplitudes and phases (Fourier); here f is the main (fundamental) frequency and $3f$ the second component.

1 pt Time domain: time on the horizontal axis and the value (strength) of the signal on the vertical axis; the drawing shows the f wave, the faster $3f$ wave (three cycles for each cycle of f) and their sum, the composite.

1 pt Frequency domain: frequency on the horizontal axis and one vertical line per sine wave at its own frequency — two lines, at f and at $3f$, with the height showing the amplitude.

1 pt Bandwidth: the range of frequencies contained in a composite signal, normally a difference between two numbers (highest minus lowest); for f and $3f$ it is $3f - f = 2f$.

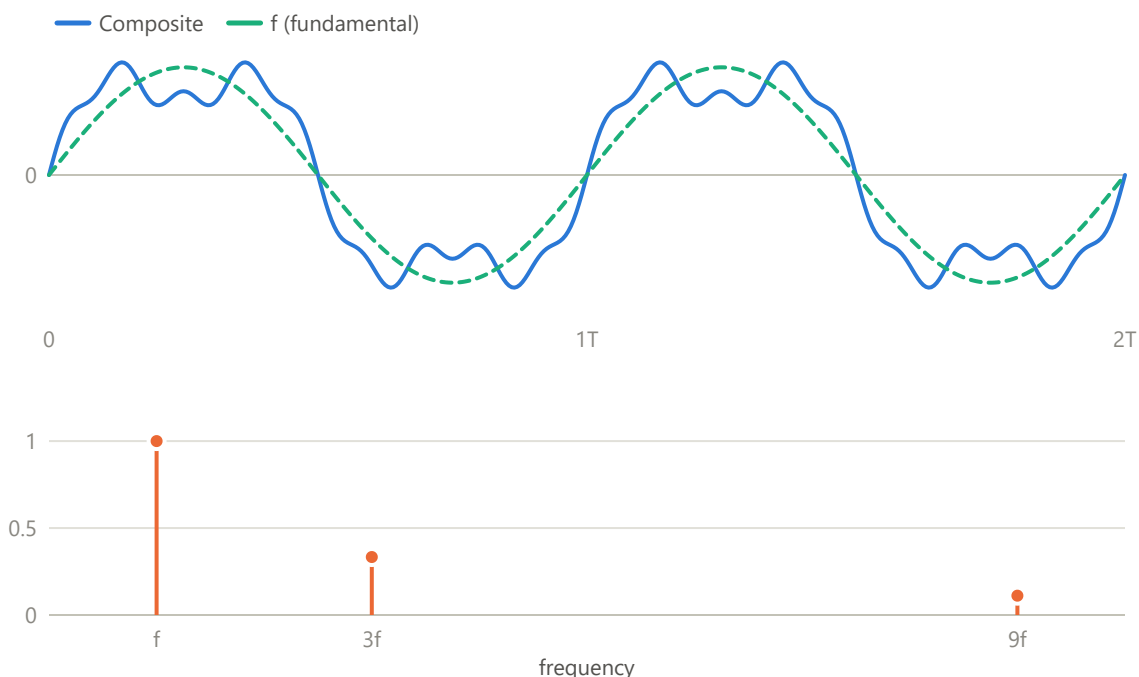
1 pt Slide example: components from 1000 Hz to 5000 Hz give $B = 5000 - 1000 = 4000$ Hz.

MODEL ANSWER

A **composite signal** is a combination of simple sine waves with different frequencies, amplitudes and phases (Fourier), and it can be periodic or nonperiodic. L02 p11 Slide 12 builds one from f , $3f$ and $9f$; the main frequency is the **fundamental frequency**. Here the signal has just f and $3f$. L02 p12

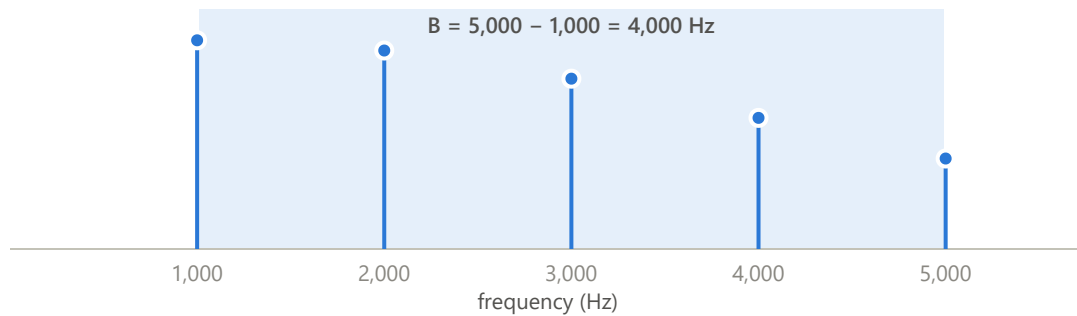
Time domain. The vertical axis is the value (strength) of the signal and the horizontal axis is time. L02 p4 Draw the f wave (one cycle in the period $T = 1/f$), L02 p7 the $3f$ wave (three faster cycles in the same time), and add them point by point to get the composite. Because $3f$ completes exactly three cycles in each period of f , the composite repeats every $T = 1/f$.

Frequency domain. The same signal drawn against frequency: each sine wave becomes one vertical line at its own frequency, and the height of the line shows its amplitude. This spectrum has two lines, one at f and one at $3f$. The time view shows the shape of the signal; the frequency view shows which frequencies it contains. L02 p12



Slide 12 uses f , $3f$ and $9f$: time domain on top, frequency domain below. For an f and $3f$ signal, leave out the $9f$ wave and the $9f$ line.

Bandwidth is the range of frequencies contained in a composite signal; it is normally a difference between two numbers, the highest frequency minus the lowest. L02 p13 For f and $3f$: $B = 3f - f = 2f$. For the slide example, a spectrum running from 1000 Hz to 5000 Hz has $B = 5000 - 1000 = \mathbf{4000\text{ Hz}}$.



Slide 13: bandwidth = highest frequency – lowest frequency.

L02 pp11-13; L02 p4; L02 p7

3. ESSAY

Compare **baseband** and **broadband** transmission. Draw a **low-pass** channel and a **bandpass** channel on a frequency axis, then explain when each kind of transmission is used. You may draw, explain, or both.

RUBRIC

1 pt Baseband: sending a digital signal over a channel without changing it into an analog signal; it needs a wide-bandwidth medium, a dedicated medium whose bandwidth constitutes only one channel.

1 pt Low-pass channel with limited bandwidth: the digital signal can only be approximated by an analog signal, and how good the approximation is depends on the bandwidth available (the first harmonic alone needs $B = N/2$; adding $3N/2$ and $5N/2$ needs $B = 5N/2$ and gives a closer match).

1 pt Broadband: the digital signal is changed into an analog signal (modulation) for transmission, which lets us use a bandpass channel, a channel with a bandwidth that does not start from zero.

1 pt When each is used: baseband when a wide-bandwidth dedicated medium is available; broadband, with a modulator at the sender and a demodulator at the receiver, when the medium has limited bandwidth and cannot pass the digital signal as it is.

1 pt A labelled drawing (or description) of the two channels on a frequency axis: the low-pass band starting at 0 Hz and the bandpass band between two frequencies above 0 Hz.

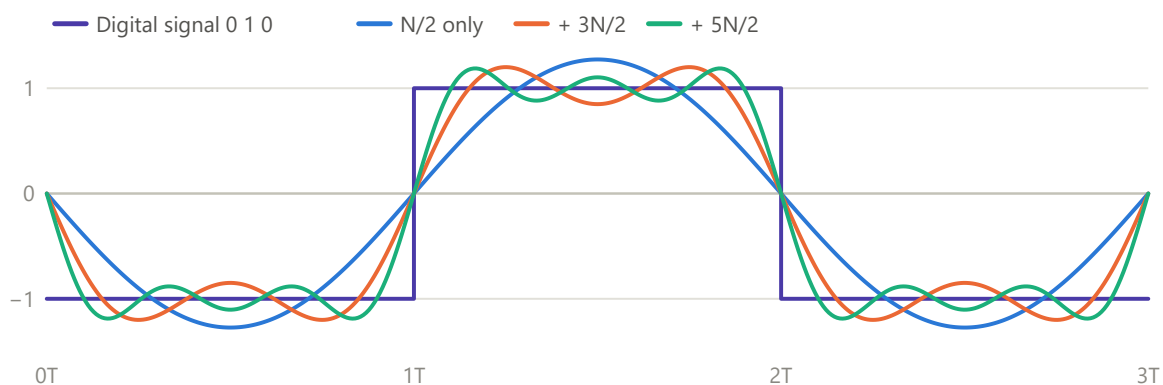
MODEL ANSWER

Baseband transmission means sending a digital signal over a channel *without* changing it into an analog signal. It is possible for a **wide-bandwidth** medium — a dedicated medium whose bandwidth constitutes only one channel.

L02 p16

A **low-pass** channel is one whose band starts at zero (a bandpass channel, below, is the kind that does not). When a low-pass channel has only limited bandwidth, the digital signal can only be **approximated** by an analog signal, and the level of approximation depends on the bandwidth available: the first harmonic alone needs $B = N/2$ (N is the bit rate), and adding the harmonics at $3N/2$ and $5N/2$ needs $B = 5N/2$ but gives a closer copy of the digital signal.

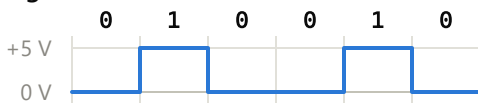
L02 p17



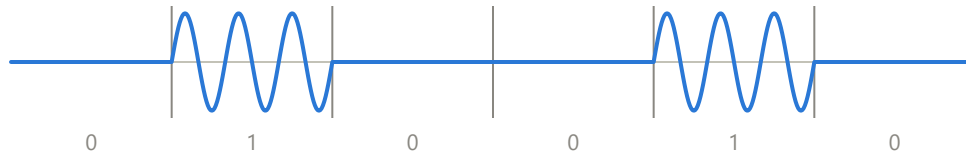
Slide 17: in a low-pass channel the digital signal is approximated by an analog one; the first harmonic needs $B = N/2$, adding $3N/2$ and $5N/2$ needs $B = 5N/2$.

Broadband transmission means changing the digital signal into an analog signal for transmission. Modulation allows us to use a **bandpass channel** — a channel with a bandwidth that does not start from zero. L02 p18

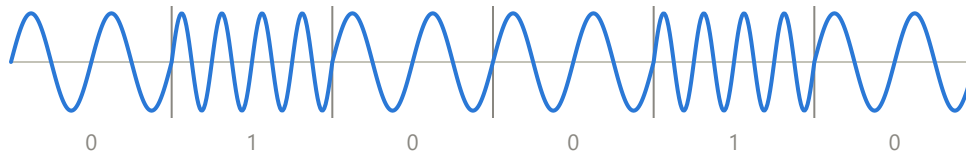
Digital transmission: 1 = +V, 0 = 0 V



Analog transmission: amplitude changes (AM) ...



... or frequency changes (FM)



Same bits, two ways: baseband sends the digital levels as they are; broadband sends an analog signal whose amplitude or frequency carries the bits (modulation).

The two channels on a frequency axis:

Channel	Band on the frequency axis	Carries
Low-pass	from 0 Hz up to an upper limit — the band starts at zero	Baseband: the digital signal itself, or an approximation of it L02 p16 L02 p17
Bandpass	between two frequencies f_1 and f_2 , both above 0 Hz — the band does not start from zero	Broadband: the modulated analog signal L02 p18

When each is used. Baseband is used when the medium is a wide-bandwidth, dedicated one. When bandwidth is limited, the medium does not let the digital signal pass; the signal is then modulated into an analog signal that a bandpass channel can carry, with a modulator at the sender and a demodulator at the receiver (one pair for each direction) — broadband transmission. [L02 p16](#) [L02 p18](#) [L01 p17](#)

[L02 pp15–18](#); [L01 p17](#)

4. ESSAY

Draw the **block coding** process for **4B/5B** and explain its three steps. Say why 4B/5B is paired with **NRZ-I**, what it does to the bit rate, and how block coding differs from **scrambling**. You may draw, explain, or both.

RUBRIC

1 pt The three steps of block coding (mB/nB): division of the bit stream into m-bit groups, substitution of each group by an n-bit group, and combination of the n-bit groups into one stream; the extra bits are redundancy that helps synchronization and error detection.

1 pt A clear labelled drawing of 4B/5B: 4-bit groups replaced by 5-bit code words from the table and joined into one stream (e.g. 0000 → 11110 and 0001 → 01001), then line-coded with NRZ-I.

1 pt Why NRZ-I: it inverts only for a 1, so a long run of 0s is a flat line with no self-synchronization; the 4B/5B code words never leave more than three 0s in a row (0000 becomes 11110), so NRZ-I keeps getting transitions while needing only N/2 bandwidth.

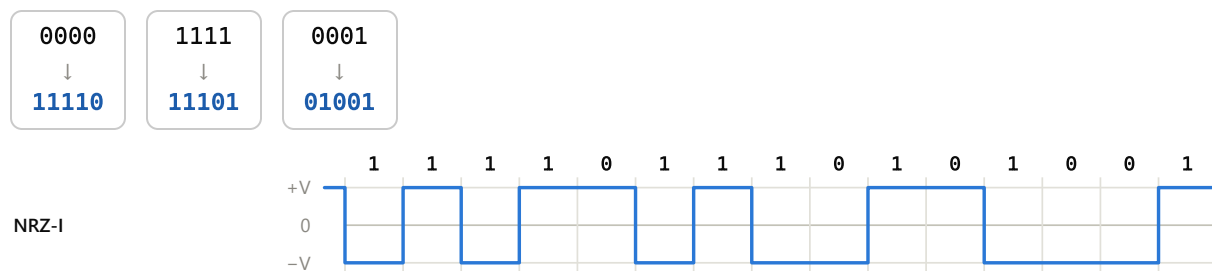
1 pt Rate: every 4 data bits become 5 code bits, so the rate rises by 5/4 (1 Mbps becomes 1.25 Mbps); NRZ-I then needs $N/2 = 625$ kHz, whereas Manchester would need 1.25 MHz (more bandwidth, but no DC component problem).

1 pt Versus scrambling: scrambling does not increase the bandwidth; it is done at the same time as the encoding, on the fly, and replaces only “unfriendly” runs of bits (e.g. B8ZS: eight 0s become 000VB0VB) with a violation code the receiver recognizes, whereas block coding substitutes every group from a table and so raises the bit rate.

MODEL ANSWER

Block coding is written **mB/nB** (with a slash) and replaces each m-bit group with an n-bit group in three steps: **division** of the bit stream into m-bit groups, **substitution** of each group by an n-bit group, and **combination** of the n-bit groups into one stream. The extra bits are **redundancy**: they let the receiver stay synchronized and detect errors, and the code words are chosen to avoid bit combinations that would cause DC components or poor synchronization under the line code. L03 p48 L03 p49

Drawing 4B/5B for the data 0000 1111 0001: divide it into three 4-bit groups; substitute each with the table (0000 → 11110, 1111 → 11101, 0001 → 01001); combine them into 11110 11101 01001 (15 bits for 12) and send the result with NRZ-I. L03 p50 L03 p51



Division into 4-bit groups → substitution with 5-bit codes → combination, then NRZ-I line coding (slide 50). The long run of 0s in 0000 becomes 11110.

Why NRZ-I. NRZ-I inverts the level for a 1 and keeps it for a 0, L03 p22 so a long run of 0s is a flat line: no self-synchronization (and a DC component problem). L03 p24 L03 p46 Reading the slide-51 table, no code word starts with more than one 0 or ends with more than two, so the coded stream never has more than three 0s in a row — even 0000 is sent as 11110 — and NRZ-I gets a transition at least every four bits, while its bandwidth stays low ($N/2$).

L03 p50 L03 p51

The cost: a higher bit rate. Four data bits become five, so the line rate is multiplied by 5/4. Example 4 (slide 54): 1 Mbps becomes 1.25 Mbps, so NRZ-I needs a minimum bandwidth of $N/2 = 625$ kHz, while Manchester would need 1.25 MHz. NRZ-I needs less bandwidth but keeps the DC component problem; Manchester needs more but has no DC problem. L03 p54 There are $2^4 = 16$ data words and $2^5 = 32$ code words, so $32 - 16 = 16$ words are left over, and some of them are used for control and signalling. L03 p53

Versus scrambling. Block coding adds extra bits to every group, so the bit rate — and with it the bandwidth the line code needs — rises by n/m . Scrambling aims for a code that does *not* increase the bandwidth for synchronization and has no DC component: it is done at the same time as the encoding, creating the bit stream on the fly, and replaces only “unfriendly” runs of bits with a violation code that is easy to recognize. L03 p57 Example: B8ZS replaces eight consecutive 0s with 000VB0VB, a pattern of the same length. L03 p59

L03 pp47–57; L03 pp22–24; L03 p46; L03 p59

5. ESSAY

Compare **delta modulation** (DM) with **PCM**. Draw a short DM staircase together with the bits it sends, say what DM sends for each sample, and explain the trade-off. You may draw, explain, or both.

RUBRIC

1 pt What DM sends: only the difference between pulses, as a single bit per sample — 1 when the signal is higher (a positive change), 0 when it is lower.

1 pt The comparison rule, in both wordings: slide 84 compares each pulse (time $t(n+1)$) with the previous pulse (time $t(n)$); the slide-85 figure, like the textbook, compares the sample with the staircase built from the bits already sent.

1 pt A staircase drawing that steps up for every 1 and down for every 0, with the bits shown (e.g. the slide-85 bits 011111100000011: one step down, six up, six down, two up).

1 pt Against PCM: PCM sends a whole code word for every sample (bit rate = bits per sample $\times$ sampling rate, e.g. 8 bits $\times$ 8000 samples/s = 64 kbps for voice), whereas DM sends 1 bit per sample, so its bit rate is just the sampling rate.

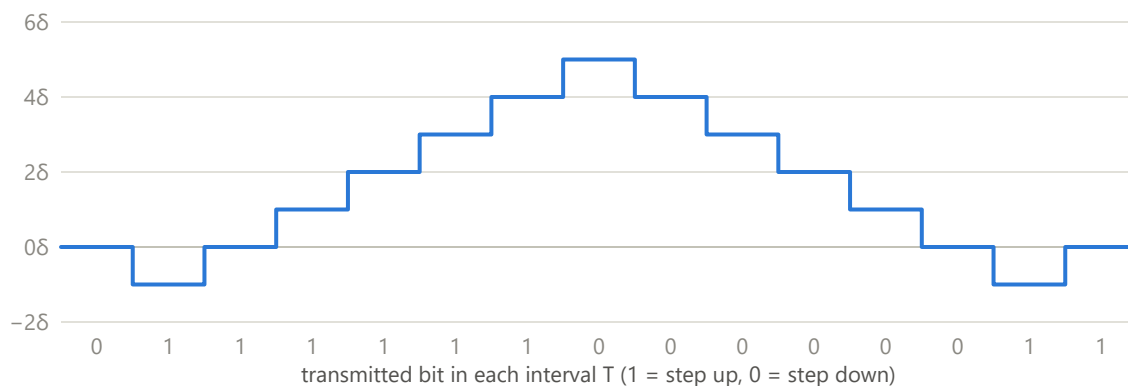
1 pt Trade-off: DM works well for small changes between samples, but large changes in amplitude cause large errors; more bits (DPCM quantizes the difference) give more levels and higher accuracy.

MODEL ANSWER

What DM sends. Delta modulation sends only the *difference* between pulses, using a **single bit per sample**: a 1 when the signal is higher than before and a 0 when it is lower. L03 p84

The rule, in both wordings. Slide 84 words it as comparing each pulse (at time t_{n+1}) with the previous pulse (at time t_n). The slide-85 figure, like the textbook, compares each sample with the **staircase**, the approximation built from the bits already sent: above it $\rightarrow$ send 1 and step up; below it $\rightarrow$ send 0 and step down. Use the staircase version when drawing. L03 p84 L03 p85

Drawing. Write the bits under a time axis and draw one step per bit — up for a 1, down for a 0. The slide-85 bits 011111100000011 give one step down, six up, six down and two up. L03 p85



Slide 85 bits 0 1 1 1 1 1 0 0 0 0 1 1 (1 = step up, 0 = step down).

Against PCM. PCM sends a whole code word for every sample: bit rate = bits per sample $\times$ sampling rate, e.g. 8 bits $\times$ 8000 samples/s = 64 kbps for digitized voice. L03 p79 L03 p80 DM sends 1 bit per sample, so with the same formula its bit rate is just the sampling rate — far fewer bits than PCM at the same sampling rate.

The trade-off. One bit per sample is economical, and DM works well when the signal changes only a little between samples. When the amplitude changes a lot, the one-bit step cannot keep up and the result is **large errors**. L03 p84 Spending more bits fixes this: DPCM quantizes the difference with several bits instead of one, so there are more levels and the accuracy is higher — at the cost of more bits per sample. L03 p88

6. ESSAY

Compare **parallel** and **serial** transmission, and the three kinds of serial transmission: **asynchronous**, **synchronous** and **isochronous**. Draw an asynchronous frame, labelling the start and stop bits. You may draw, explain, or both.

RUBRIC

- 1 pt** Parallel: multiple bits are sent with each clock tick; serial: 1 bit is sent with each clock tick, in three subclasses — asynchronous, synchronous and isochronous.
- 1 pt** Asynchronous: 1 start bit (0) at the beginning and 1 or more stop bits (1s) at the end of each byte, with possible gaps between bytes; it is asynchronous only at the byte level, since the bits themselves are still synchronized (same duration).
- 1 pt** Synchronous: bits are sent one after another without start or stop bits or gaps; the receiver must group the bits, and many bytes are grouped in a frame identified by a start byte and an end byte.
- 1 pt** Isochronous: no uneven gaps between frames; the transmission of bits is fixed, with equal gaps.
- 1 pt** A labelled drawing of an asynchronous frame: start bit (0) first, then the data byte, then the stop bit(s) (1), with a gap before the next byte.

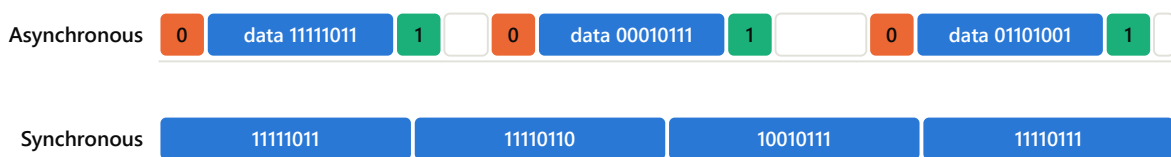
MODEL ANSWER

Parallel vs serial. In **parallel** mode multiple bits are sent with each clock tick; in **serial** mode 1 bit is sent with each clock tick. Serial transmission has three subclasses: **asynchronous**, **synchronous** and **isochronous**. L03 p89

Beyond slides In the textbook, parallel mode uses n wires to send n bits at once: fast, but costly, so it is usually limited to short distances. Serial mode needs only one channel, with a parallel-to-serial converter at the sender and a serial-to-parallel converter at the receiver.

Asynchronous. Send **1 start bit (0)** at the beginning and **1 or more stop bits (1s)** at the end of each byte; there may be a **gap** between bytes. It is “asynchronous at the byte level”, but the bits are still synchronized and last the same time. The start and stop bits add at least two extra bits to every byte. L03 p92

Drawing an asynchronous frame. Draw the start bit (0) first, then the eight data bits, then the stop bit (1) — more are allowed — and leave a gap before the next byte. Slide 93 draws each byte as [stop 1 | data | start 0] with the start bit first on the line, so on a left-to-right time axis the order is start 0, data, stop 1. L03 p93



0 start bit (0) 1 data byte 1 stop bit (1)

Top: asynchronous, with a start bit (0) in front of every byte, a stop bit (1) behind it and gaps allowed between bytes. Bottom: synchronous, with bytes back to back in a frame. Schematic only: in a real signal every bit lasts the same time.

Synchronous. Bits are sent one after another **without start or stop bits or gaps**; it is the receiver's responsibility to group the bits. They are usually sent as bytes, and many bytes are grouped in a **frame** that is identified by a start byte and an end byte. L03 p94

Isochronous. There cannot be uneven gaps between frames: the transmission of bits is fixed, with equal gaps.

L03 p96 **Beyond slides** The textbook uses it for real-time audio and video, where uneven delays between frames are not acceptable.

7. ESSAY

Draw the **QPSK** constellation with the slides' bit labels and explain what a constellation diagram shows. Then explain how **16-QAM** carries more bits per signal element.

RUBRIC

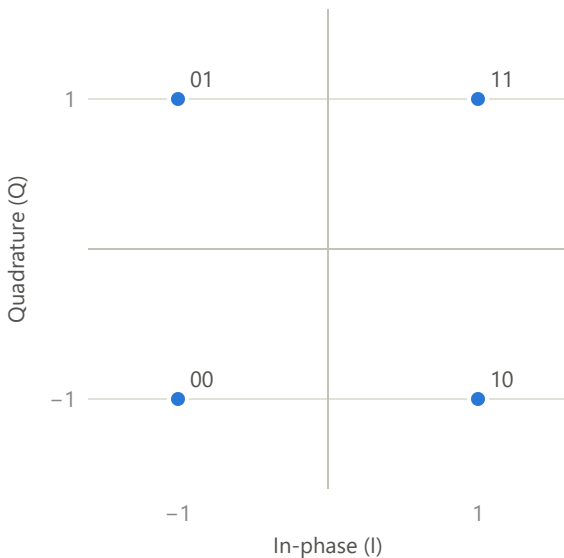
- 1 pt** Axes: in-phase (I) horizontal, quadrature (Q) vertical; each point is one signal element.
- 1 pt** Distance from the origin = amplitude; angle = phase.
- 1 pt** QPSK: four points of equal amplitude at 45° , 135° , -135° and -45° , labelled 11, 01, 00, 10 (the first bit gives the I sign, the second the Q sign).
- 1 pt** QPSK carries $r = 2$ bits per element ($L = 4$), so $S = N/2$.
- 1 pt** 16-QAM changes both amplitude and phase: 16 points, $r = 4$ bits per element, so $S = N/4$ — more bits per element at the same baud rate.

MODEL ANSWER

A constellation diagram plots each signal element as a point: the in-phase carrier on the horizontal axis and the quadrature carrier on the vertical axis. The point's distance from the origin is the amplitude and its angle is the phase.

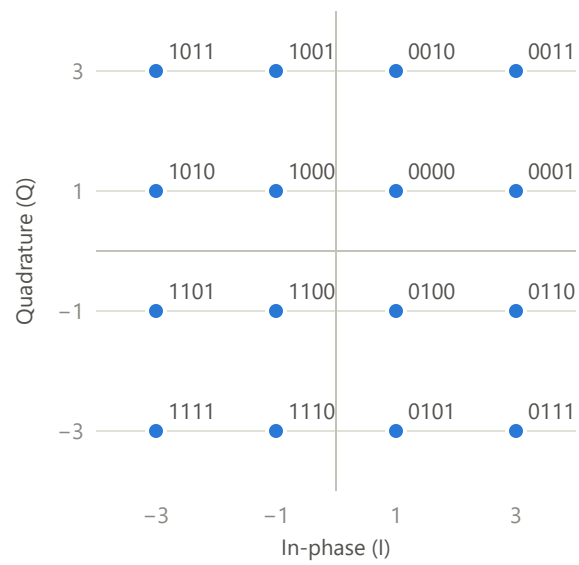
L04 p27

L04 p28



QPSK sends two bits per element: the first bit sets the I sign and the second the Q sign, giving four equal-amplitude points — 11 at 45° , 01 at 135° , 00 at -135° , 10 at -45° . With $r = 2$, $S = N/2$. L04 p25 L04 p29

16-QAM combines ASK and PSK: points differ in amplitude and phase, so there are 16 of them and each carries $r = 4$ bits — $S = N/4$. L04 p30 L04 p31



L04 pp24-31

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